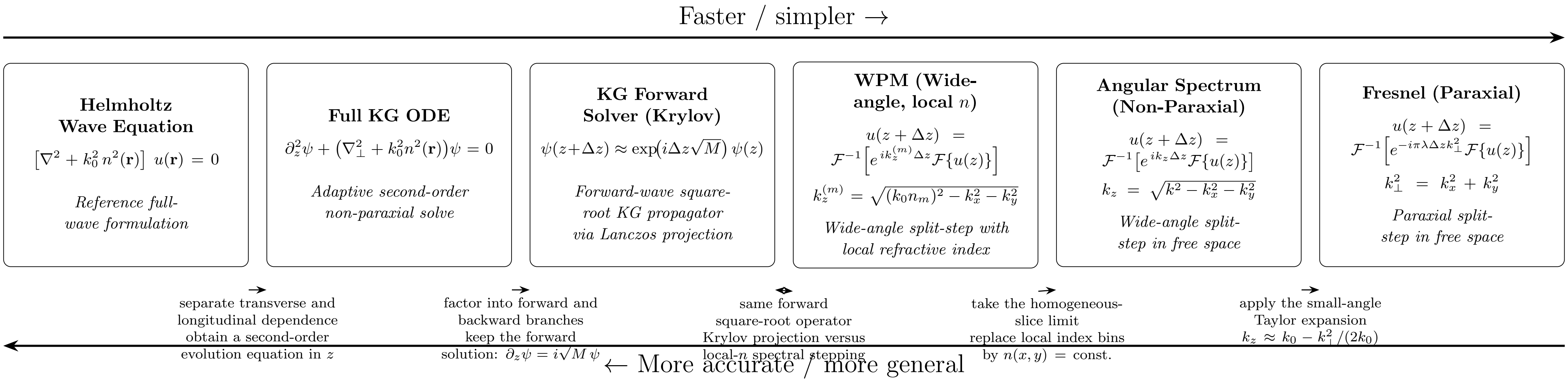
Faster / simpler →
Helmholtz Wave Equation
$\left[\nabla^2 + k_0^2\, n^2(\mathbf{r})\right]\; u(\mathbf{r}) \,=\, 0$
Reference full-wave formulation
Full KG ODE
$\partial_z^2 \psi + \left(\nabla_\perp^2 + k_0^2 n^2(\mathbf{r})\right)\psi = 0$
Adaptive second-order non-paraxial solve
KG Forward Solver (Krylov)
$\psi(z{+}\Delta z) \approx \exp\!\left(i\Delta z\sqrt{M}\right)\psi(z)$
Forward-wave square-root KG propagator via Lanczos projection
WPM (Wide-angle, local n)
$u(z+\Delta z) \;=\; \mathcal{F}^{-1}\!\left[e^{\,i k_z^{(m)}\Delta z}\mathcal{F}\{u(z)\}\right]$
$k_z^{(m)} \;=\; \sqrt{(k_0 n_m)^2 - k_x^2 - k_y^2}$
Wide-angle split-step with local refractive index
Angular Spectrum (Non-Paraxial)
$u(z+\Delta z) \;=\; \mathcal{F}^{-1}\!\left[e^{\,i k_z \Delta z}\mathcal{F}\{u(z)\}\right]$
$k_z \;=\; \sqrt{k^2 - k_x^2 - k_y^2}$
Wide-angle split-step in free space
Fresnel (Paraxial)
$u(z+\Delta z) \;=\; \mathcal{F}^{-1}\!\left[e^{-i\pi\lambda\Delta z k_\perp^2}\mathcal{F}\{u(z)\}\right]$
$k_\perp^2 \;=\; k_x^2 \,+\, k_y^2$
Paraxial split-step in free space
separate transverse and longitudinal dependence
obtain a second-order evolution equation in z
factor into forward and backward branches
keep the forward solution: $\partial_z\psi = i\sqrt{M}\,\psi$
same forward square-root operator
Krylov projection versus local-n spectral stepping
take the homogeneous-slice limit
replace local index bins by $n(x,y) \,=\,$ const.
apply the small-angle Taylor expansion
$k_z \,\approx\, k_0 - k_\perp^2/(2k_0)$
← More accurate / more general